

PAPER

Bioaccumulation and toxicity of 2-naphthalene sulfonate: an intermediate compound used in textile industry

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Abstract

The aromatic compounds substituted with sulfonate groups, being xenobiotic, resist biodegradation in the environment and tend to accumulate up to toxic levels. The hydrophilic sulfonated group makes these compounds highly water soluble and they tend to pass through water-treatment plants. The release of untreated effluents from these industries results in pollution of water bodies affecting aquatic fauna. Thus, the toxicity regarding these compounds is of major concern. The 2-naphthalene sulfonate is a sulfonated aromatic compound being widely used in textile industries. Being non-biodegradable concern regarding its toxicity has risen. Thus in the light of above facts, the present study was undertaken to determine the toxicity of 2-naphthalene sulfonate in blood cells of *Channa punctatus*. For this, LD₅₀ was determined and after selection of sublethal doses oxidative stress, genotoxicity and bioaccumulation were studied. For oxidative stress determination, biochemical markers such as malondialdehyde content and activities of superoxide dismutase, catalase, and glutathione-S-transferase were studied. Genotoxicity was studied using comet and micronucleus assay. Significant increase in oxidative stress and DNA damage in the exposed groups as compared to control group ($P \leq 0.05$) was observed till 96 h. However, decreased values of all the studied parameters at 720 h (30 days) indicate repair capacity of fish. Further, the bioaccumulative potential of 2-naphthalene sulfonate was assessed in blood plasma using high-performance liquid chromatography. The study revealed the toxic potential of 2-naphthalene sulfonate to aquatic organisms thus stressed on the need for the implementation of stringent policies regarding the management of such toxic compounds.

Key words: 2-naphthalene sulfonate; LD₅₀; oxidative stress; genotoxicity; bioaccumulation

Introduction

The increasing industrialization and urbanization has resulted in rapid elevation in environmental pollution. The discharge of untreated solid and liquid wastes to the environment is one of the major problems around the globe, as it is affecting water resources, soil fertility, aquatic fauna and flora, and ecosystem stability [1–3]. Among various industries, textile industries

majorly contribute to the generation of water pollution as consumption rate of water is fairly high. For the production of 1 kg of textile, 200 l of water is consumed and as a result a huge amount of liquid waste prior to treatment is being discharged to aquatic bodies [4]. Textile industry uses a vast range of dyes, during different stages of textile processing [5]. Studies revealed that annually 280 000 tons of textile dyes are discharged in industrial effluent, out of that azo dyes make up about a half of all known dyestuffs

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in the world [6]. Textile dyes and their intermediate products are among the most prevalent and persistent pollutants in the environment. The release and accumulation of these compounds impose serious threat to public due to their carcinogenic, mutagenic, antiestrogenic, and teratogenicity potential [7–9]. Thus, textile wastewater release is directly or indirectly related with various human illnesses and environmental pollution [10].

2-Naphthalene sulfonate (2 NS) is also one of the intermediates involved in the synthesis of dyes, surfactants, and dispersants. It is widely used in concrete finishing, tanning of hides, and in the production of pharmaceuticals and agrochemicals [11–13]. It is specifically produced from the condensation process of naphthalenesulfonic acid and formaldehyde (CNSF) and is mainly used as retan agent. The sulfonic group imparts maximum stability to the benzene ring making the compound highly resistant to biodegradation [14], thus gets accumulated in aquatic bodies and further gets involved in food chain. Number of studies revealed the toxic effects of dyes and surfactants being used in textile industries [15, 16], but scarce studies are available on the toxicity induced by the intermediates. Keeping in mind, the present investigation has been carried out to study the toxic effects of 2NS in fresh water fish *Channa punctatus* using biochemical and genotoxicity markers. Further bioaccumulation of compound was also studied in blood plasma of fish using Semi Prep high-performance liquid chromatography (HPLC).

Materials and Methods

Experimental model

Fresh water fishes *C. punctatus* were procured from local fish market having weight 15 ± 2 g and measured 12 ± 2 cm. Fish was treated with 0.02% KMnO_4 for 2 min to avoid any dermal infection. The fishes were kept in rectangular glass aquaria of capacity 200 l and acclimatized for 15 days under laboratory in static conditions in laboratory. They were fed with synthetic fish diet. The fecal matter and other waste materials were siphoned off daily to reduce ammonia content in water.

2NS was purchased from Himedia Research laboratory, Mumbai, India (CAS No. 120-18-3). The stock solution of 2NS was prepared by dissolving 1 g of 2NS in 10 ml of distilled water; further dilutions were prepared from this stock solution.

LD₅₀ determination and behavioral study

Preliminary tests were performed to determine appropriate range of toxicity of 2-naphthalene sulfonate. Fishes were divided into batches of 10 each and were exposed to different concentrations of 2NS ranging from 0.2 to 1 mg/15 g (body weight [b.w.]). The doses were injected intraperitoneally midline between the pelvic fins. Mortality of fish was recorded at 24, 48, 72, and 96 h of exposure with 2NS. The percentage mortality of *C. punctatus* was observed to be 0 and 100% at 0.2 and 1.0 mg/15 g (b.w.) concentration of 2NS, respectively. The experiments were performed in duplicates and repeated thrice to confirm the results. LD₅₀ value was calculated by using computer software Probit Analysis [17] and behavior alterations were observed.

Experimental design

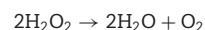
After LD₅₀ determination, two concentrations, i.e. 0.16 (one-fourth of LD₅₀) and 0.33 mg/15 g (b.w.) (one-half of LD₅₀) were selected for biochemical and genotoxicity investigation and

further bioaccumulation studies were done in the plasma of fish exposed to 0.33 mg/15 g (b.w.). For biochemical investigation, different parameters like MDA, SOD, CAT, and glutathione-S-transferase (GST) activities were assessed and genotoxicity was determined using micronucleus and comet assay. The experiment was performed up to 720 h (30 days) and the samples were taken after 24, 48, 72, 96, 240, 480, and 720 h of exposure.

Biochemical parameters

MDA. The lipid peroxidation was determined according the method of Okhawa [18]. The method is based on the formation of thiobarbituric acid (TBA) reactive metabolites of lipids such as malondialdehyde (MDA). Degradation products of peroxidized lipids form MDA, which forms adduct with TBA. MDA-TBA chromophore formed is measured calorimetrically at 532 nm.

Catalase. Catalase activity was evaluated by the method of Aebi [19]. The catalase is the enzyme which helps in catalyzing decomposition of hydrogen peroxide and leads to a decrease in its UV absorption with time and this decrease in absorbance is the measure of enzyme activity.



Formula used

$$\text{Activity} = \frac{\Delta E / \text{min} * \text{total volume of reaction}}{\text{EC} * \text{sample volume} * \text{protein concentration}}$$

$$\text{EC} = 71 (\text{M cm})^{-1}$$

Result obtained was in n-moles of H_2O_2 decomposed/min/mg protein.

Glutathione-S-transferase. GST activity was estimated according to the Habig [20]. In this method, 1-chloro-2,4-dinitro-benzene (CDNB) was used as substrate. GST enzyme is involved in catalyzing the formation of conjugate between reduced glutathione (GSH) and CDNB. The OD was read at 340 nm.

Formula used

$$\text{Activity} = \frac{\Delta \text{OD} / \text{min} * \text{total vol}}{\text{EC} * \text{volume of sample} * \text{protein (mg/ml)}}$$

$$\text{EC} = 9.6 * 10^6 (\text{M cm})^{-1}$$

Result was obtained as n-moles of CDNB-GSH conjugate formed/min/mg protein.

Superoxide dismutase. Superoxide dismutase (SOD) activity was determined according to the method of Kono [21]. Superoxide anions are generated by the oxidation of hydroxylamine hydrochloride. The blue formazon was formed after the reduction of nitroblue tetrazolium (NBT) by superoxide anions. SOD inhibits the reduction of NBT mediated by hydroxylamine hydrochloride, and the extent of inhibition is taken as a measure of enzyme activity recorded at 560 nm.

Formula used

$$X (\% \text{Inhibition}) = \frac{\Delta \text{non-enzymatic} - \Delta \text{enzymatic} * 100}{\Delta \text{non-enzymatic}}$$

$$\text{Units of SOD/mg protein} = \frac{X}{\text{protein conc} \times \text{sample vol}}$$

Estimation of protein content

Protein content was estimated according to Bradford assay using BSA as standard [22].

Genotoxicity

For genotoxicity, testing micronucleus assay was performed according to methodology of Kumar [23] and comet assay using the method of [24].

Micronucleus assay. For micronucleus assay, blood smear was prepared by placing a drop of blood on a clean slide and dried at room temperature for experimental as well as control groups. Slides were then fixed in fixative solution for 10 min and stained with 10% Giemsa. The dried slides were finally analyzed to evaluate and quantify the presence of micronucleated cell (MNC) and aberrant cell (AC) frequency. For each group (treated as well as control), 1000 erythrocytes were counted for studying micronucleated and AC frequency. Nuclear abnormalities include budding, blebbing, notching of nucleus, nuclear bridge, fragmented nucleus, deformed nuclei, and vacuolated nucleus and cytoplasmic abnormalities include vacuolated cytoplasm, cytolysis, cytoplasmic bridge, and swelled cells. All these cells with cytoplasmic and nuclear aberrations are collectively considered as AC.

Comet assay. The alkaline SCGE was performed using method of Ahuja and Saran [24] using the blood sample taken from the heart. Blood was immediately diluted in phosphate saline buffer (10 ml blood in 1 ml of PBS). Microscope slides were coated with 1% normal melting point agarose and were incubated at 37°C overnight. To the coated slides, 85 ml of low melting point agarose (LMPA 0.5%) was added and mixed with 20 ml of blood. The slides were covered with a cover slip and placed on a slide tray in a refrigerator at 4°C for 10 min. Finally, a third layer of LMPA (0.5%) was poured on the slide and returned to the refrigerator at 4°C for another 10 min. Then the slides were placed in lysis buffer for about 3 h in the refrigerator. After lysis, the slides were incubated for 20 min in electrophoretic buffer followed by electrophoresis for 20 min at 300 mA and 24 V. The slides were then neutralized with the neutralization buffer for 15 min. After overnight drying, the slides were stained with ethidium bromide and analyzed under a fluorescence microscope.

Bioaccumulation investigation

Shimadzu Prep HPLC System, basically used for analytical and preparative purpose, equipped with fraction collector, and UV/Vis detector was used for the present analysis, with C₁₈ column, acetonitrile, and water as mobile phase in the ratio of 30:70, and the flow rate of 1 ml/min. The column was maintained at 30°C. Prior to use, solvents chosen as mobile phase were filtered through Millipore 0.45 µm. The detection wavelength of the detector was set at 270 nm. For HPLC analysis, stock and standard solutions were prepared by dissolving an appropriate amount of compound in order to obtain concentration of 1 mg/1 ml. Working standard solutions were prepared by gradual dilutions each day with the mobile phase. Standard curves were plotted

and using peak areas of known concentrations, quantification of 2NS in plasma samples was performed.

Sample preparation. In order to precipitate proteins, 3 ml of acetonitrile was added to 1 ml of plasma. Further, the mixture was vortexed for 1 min, left overnight, and centrifuged at 2000 g for 10 min. Upper layer was separated in a tube and back extracted with 200 µl of 0.3% orthophosphoric acid. The organic layer was aspirated and the 100 µl of the residue dissolved in mobile phase to make it 1.5 ml which was injected to the column with 10 µl of injection volume. Peak identification was made by comparing retention times of samples with those of the standard solutions.

Specificity of retention time for 2NS and accuracy of the method. Three replicates from the same sample of plasma were prepared, which were subjected to extraction described above. Exposed plasma sample was run along with blank and standard solution. The peak area versus concentration data were used for the further analysis. The accuracy of the method was evaluated by studying the fortified plasma sample for known concentration of 2NS in triplicate.

Statistical analysis

Statistical analysis was performed using SPSS 16.0. All data were presented as mean ± SE. One-way ANOVA followed by post-hoc Tukey's test was used to study the significant difference between control and treated groups.

Results

LD₅₀ determination and behavioral study

The LD₅₀ value for 2NS was determined to be 0.66 mg/15 g (b.w.) (Table 1). The alterations in behavior of *C. punctatus* after the intraperitoneal administration of 2NS are presented in Table 2. Abnormal behavior was observed in treated group of fishes as compared to control group. Within few minutes of exposure, disruption in schooling behavior and hyper excitability was observed. The exposed fish showed erratic swimming and equilibrium loss and spreading of an excess of mucus all over the surface of the body. After exposure to increasing concentrations of 2NS, the fishes were observed to show dark colored pigmentation (Fig. 1). After few hours, fishes became hypoactive and get submerged at the bottom of the tank prior to death. The behavior changes were found to be more pronounced after treatment with higher concentration as compared to lower concentration as shown in Table 2.

Biochemical investigation

The toxic effects of 2NS were observed by measuring oxidative stress and DNA damage in blood cells of fish after treatment with both concentrations at different time intervals. Acute effect was studied up to 96 h of duration in exposed fishes and further repair potential was studied up to 720 h (30 days). Results are presented in Fig. 2A–D. Figure 2A reveals the effect of 2NS on MDA that is found to be significantly different as compared to control after 24, 48, 72, 96, 240, and 480 h of exposure after treating with both the concentrations. Maximum effect was observed after 96 h of exposure, however, after 240 h (10 days) of exposure, the toxic effect gets reversed and after 720 h of exposure no

Table 1: Showing LD₅₀ value of 2NS

Chemical name	Animal model	LD ₅₀ (mg/15 g [b.w.])	Upper limit	Lower limit
2-Naphthalene sulfonate	<i>C. punctatus</i>	0.66	8.14	4.77

Table 2: Showing the changes in behavioral parameters after exposure of different concentrations of 2NS

Parameters	Control	0.2 mg/15 g (b.w.)	0.4 mg/15 g (b.w.)	0.6 mg/15 g (b.w.)	0.8 mg/15 g (b.w.)	0.9 mg/15 g (b.w.)	1 mg/15 g (b.w.)
Hyperactivity	–	+	+	++	+++	+++	+++
Loss of balance	–	+	+	++	+++	+++	+++
Rate of swimming	+	+	+	++	+++	+++	+++
Pigmentation	–	+	+	++	+++	+++	+++

(–) None, (+) mild, (++) moderate, and (+++) strong.

**Figure 1:** Pigmentation in fishes exposed to different concentrations of 2NS.

significant difference from control was observed indicating the repair potential of fish.

Along with MDA, specific activities of antioxidant enzymes (GST, CAT, and SOD) were assessed. In case of GST, significant increase was observed till 240 h of exposure with maximum increase at 96 h of exposure showing 31.37 and 25.98% increase after the administration of 0.33 and 0.16 mg/15 g (b.w.), respectively (Fig 2B). However, no significant difference between control and treated group was determined at 720 h (30 days) of exposure showing that the fish has undergone recovery at this stage.

Figure 2C and D depicts the enzyme activities of CAT and SOD. Decrease in the activities of these enzymes was observed as compared to control. Maximum decline was observed after 96 h of treatment showing 63.3 and 46.89% decrease in CAT and SOD activity respectively as compared to control. After 240 h of exposure, the values were found to be increased.

Genotoxicity

The DNA damage was assessed by micronucleus assay and comet assay.

Micronucleus assay. Results reveal a significant increase in MNC frequency and AC frequency as compared to control after treating with different concentrations at same interval of time (Fig. 3A and B). A dose-dependent increase in MNC frequency was observed after 96 h of exposure. The values show significant increase even after 240 h of exposure but after 480 h, sudden decrease in the values was observed revealing the reversal toxicity effect.

The highest effect was observed at 240 h of exposure. The mean MNC frequency comes out to be the highest (1.63 ± 0.24) after 240 h of exposure with 0.33 mg/15 g (b.w.). Least effect

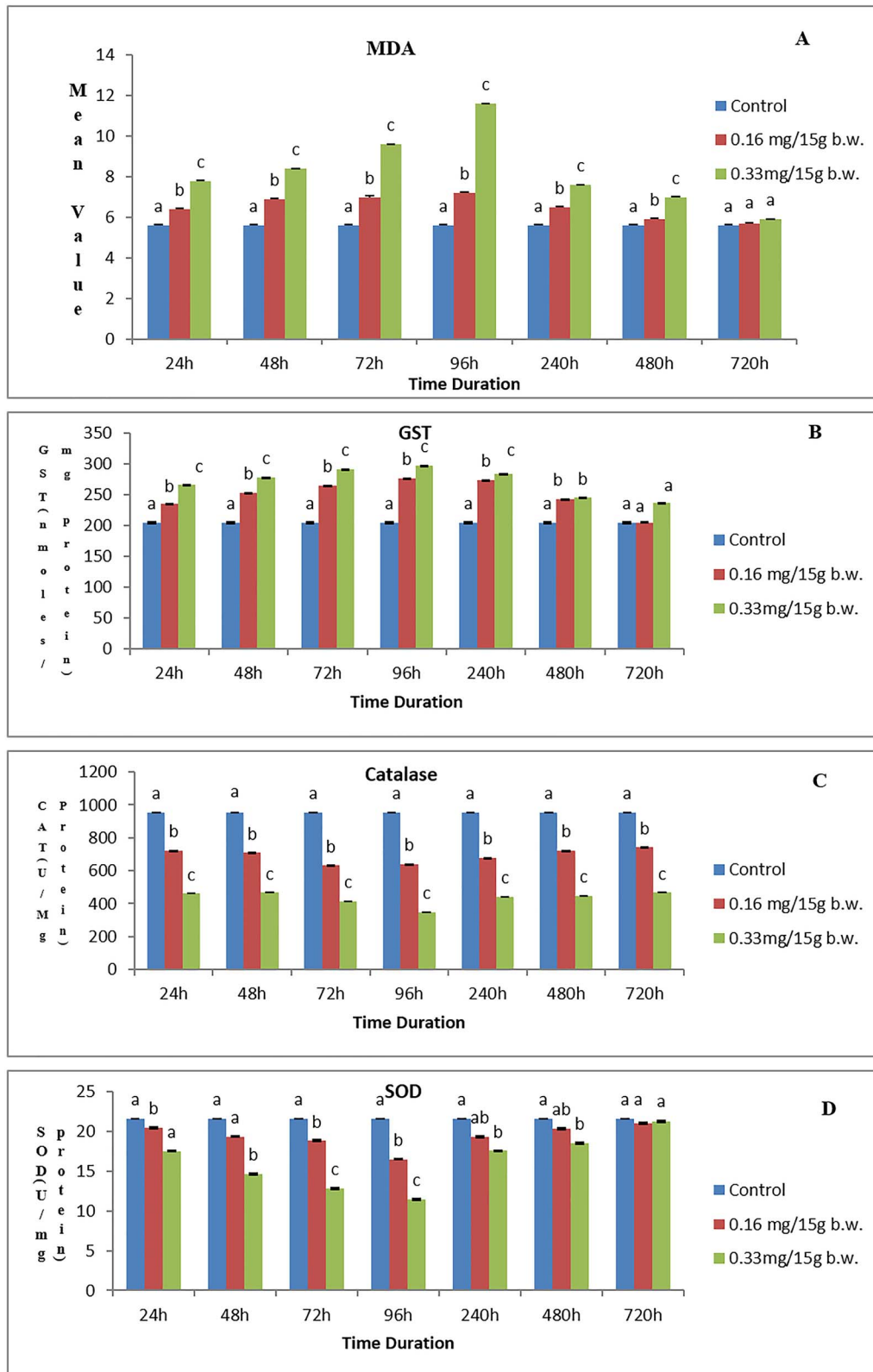


Figure 2: (A–D) Effect of 2NS on MDA content and enzyme activities in blood of *C. punctatus*. Error bars represent standard errors (SE). Different letters a–c show significant difference between different concentrations at same time interval.

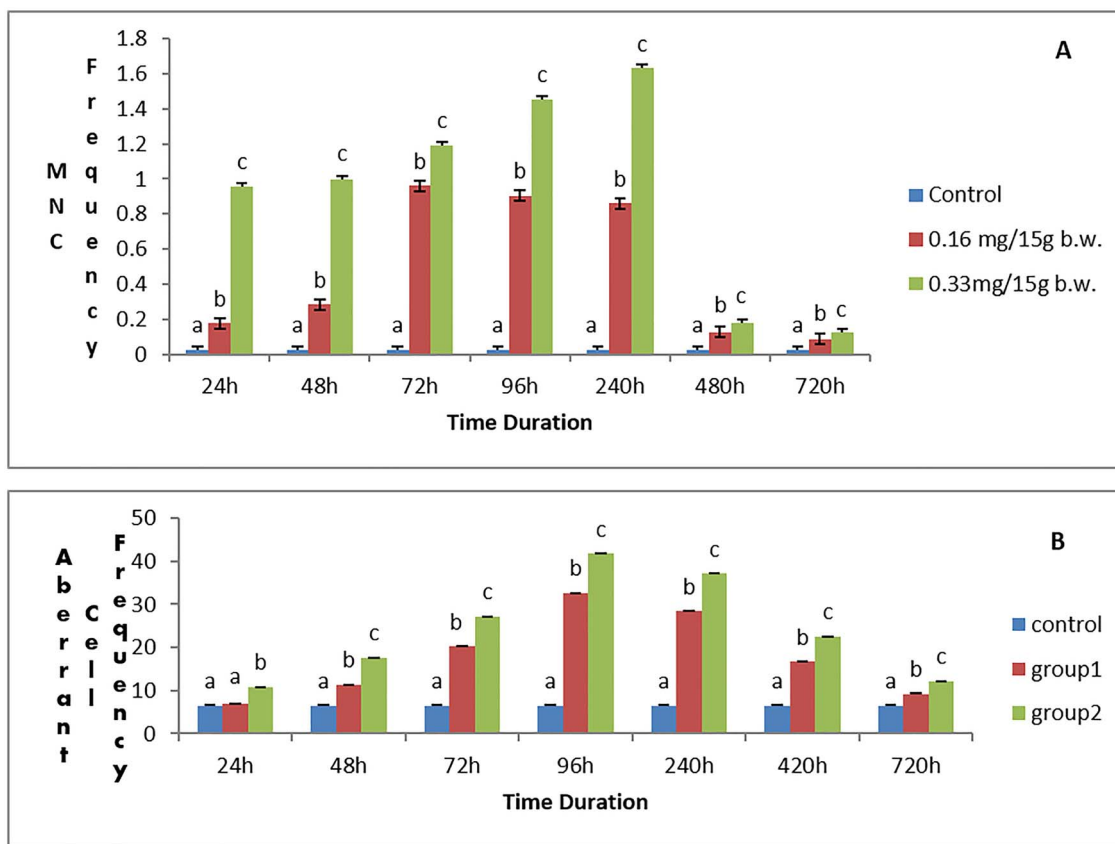


Figure 3: (A and B) Effect of 2NS on micronucleus frequency and AC frequency in blood cells of *C. punctatus*. Error bars represent SE. Different letters a–c show significant difference between different concentrations at same time interval.

was found at 720 h of administration of 2NS intraperitoneally. Similarly, a dose-dependent increase in AC frequency was determined. The highest mean AC frequency was found to be 41.73 ± 0.22 at 96 h of exposure after treatment with highest concentration. An increase in value from 6.55 (control) to 41.73% is 6.37 times more than the control.

Comet assay. The results of comet assay have been given in Fig. 4A and B.

Tail length and % tail intensity were considered as parameters for assessing the DNA damage. Significant increase in the values of both the parameters was observed at 24 h of exposure. At 48 h, a slight decrease in the value of % tail intensity was seen. Further at 72 and 96 h of treatment, the values again increased for both the parameters. Maximum damage was observed after 96 h showing 68.17% increase in tail length. The % tail intensity gives a peak shoot at 96 h with the rise from 1.60 ± 0.15 to $13.52 \pm 0.65\%$. However at 720 h of exposure, no significant difference as compared to control was observed thus verifying recovery at this stage.

Bioaccumulation of 2NS

Specificity of retention time for 2NS. The retention time was found to be 8.256 min and the detector wavelength was 270 nm. Figure 5 represents that the calibration plot for the assay of 2NS was linear over the investigated range, i.e. 5–150 $\mu\text{g/ml}$. The correlation coefficient was found to be 0.99 ($P < 0.05$). The accuracy of the method was obtained by calculating percentage recovery,

which was calculated from the concentration of the standard injected, and the concentration of standard obtained after extraction which was found to be 89.7% for 50 μg . Limit of quantification is found to be 1.83 $\mu\text{g/ml}$ and limit of detection comes out to be 0.60 $\mu\text{g/ml}$.

Concentration of 2NS in exposed fishes. Figure 6 represents the mean concentration of 2NS found in fishes exposed to 0.33 mg/15 g (b.w.) for 24, 96, 240, 480, and 720 h. Maximum accumulation was observed at 24 h of exposure. Consistently significant ($P \leq 0.05$) decrease in accumulation was observed with increase in time.

Discussion

To study the effects of water-borne contaminants, fishes are most reliable experimental model. Being the first recipient they can metabolize, concentrate, and store those contaminants in their bodies as higher vertebrates. *C. punctatus*, being widely distributed and available throughout the year, is considered as an excellent model species [25]. Present study reveals the toxicity induced in *C. punctatus* after treating with two sublethal concentrations of 2NS. LD₅₀ value of 2NS was found to be 0.66 mg/15 g (b.w.).

Change in behavior pattern is the sensitive parameter to assess the stress induced in the organisms due to environmental pollution. Different compounds are found to induce alterations in the behavior of organisms [26]. During the present investigation, changes in behavior pattern were studied after injecting 2NS intraperitoneally. Irregular movements, reduced

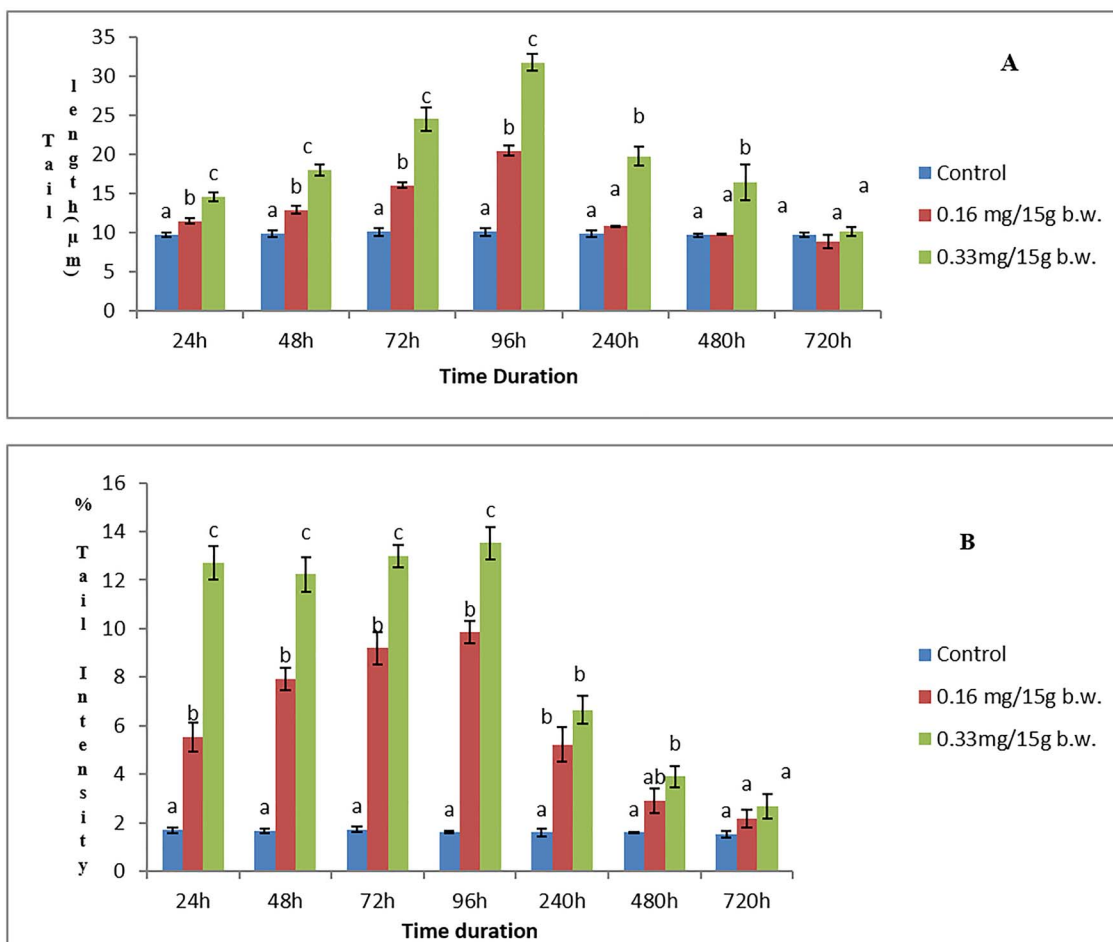


Figure 4: (A and B) Effect of 2NS on tail length and % tail intensity in blood cells of *C. punctatus*. Error bars represent SE. Different letters a-c show significant difference between different concentrations at same time interval.

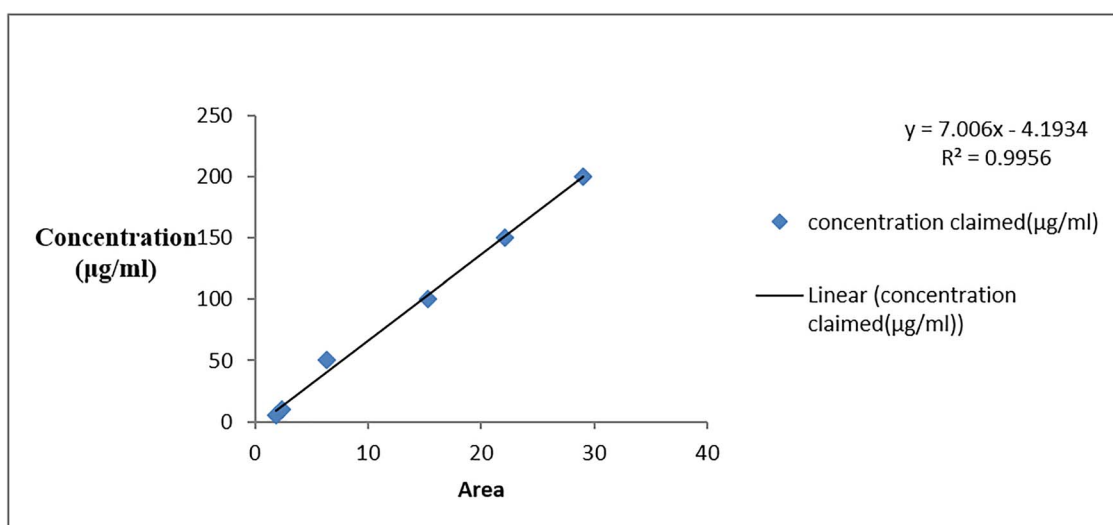


Figure 5: Calibration plot for the assay of 2NS.

reflexes, high pigmentation, and mucus secretion were observed in exposed fish. Finally, the fish get submerged at the bottom of the tank prior to death. Pathan *et al.* [27] revealed the change in behavior in *C. punctatus* after the exposure to industrial effluent. Similarly, Adeogun *et al.* [28] revealed the ill effects of textile

factory effluents on otolith and somatic growth parameters in *Clarias gariepinus*. Hussain *et al.* [29] also revealed change in pigmentation, high mucus secretion, and change in feeding pattern in *Catla catla* after the exposure of dimethoate. Similarly, studies done by Kumar *et al.* [30] showed change in rate of gulping air,

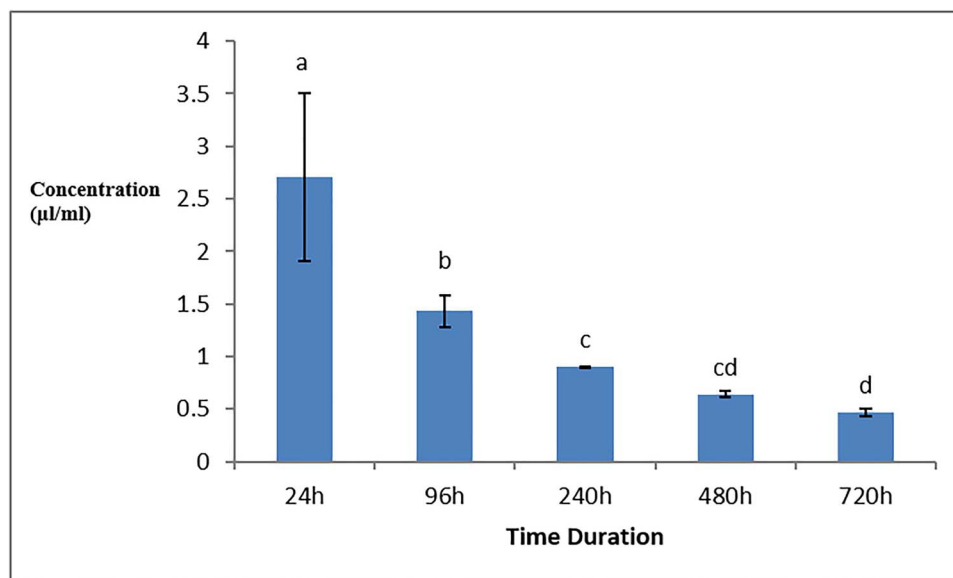


Figure 6: Mean concentration of 2NS in the blood plasma of exposed fishes at different time interval. Different letters a–d show significant difference at different time interval (one-way ANOVA).

increase in mucus secretion, muscle fasciculation, and irregular swimming patterns in fish *Clarias batrachus* after exposure to copper sulfate.

Further, the effect of sublethal concentrations after acute exposure as well as repair potential of fish *C. punctatus* was studied. For acute effect, the experiment was performed continuously till 96 h and maximum effect was observed at 96 h. Significant increase in MDA was observed. Similar results were observed by Sharma et al. [31] in *C. punctatus* after acute exposure of TBBPA. However, no significant difference between control and treated group was determined at 720 h (30 days) of exposure showing that the fish undergoes recovery at this stage. Xenobiotic affects redox balance resulting in generation of ROS inducing oxidative stress [32]. Oxidative stress may lead to damage in cellular constituents, which further led to imbalance in antioxidant defense mechanisms. Hence, in the present study, activities of antioxidant enzymes (SOD, CAT and GST) along with MDA content were revealed. GST is a vital enzyme that aids in detoxification. Significant increase in GST activity observed in the present study shows the apparent protective behavior of the enzyme against ROS production. 45.54% increase was observed at 96 h for highest dose and 35.10% for lowest dose. Reduction in values of GST activity was observed after 240 and 480 h and values get lower up to control level after 720 h. GST catalyzes the linking of xenobiotic with glutathione, thus forming less reactive substance. Besides GST, alterations in CAT and SOD were also observed. Reduction in CAT and SOD activity was found after acute exposure of 2NS for 96 h of duration for both the concentrations. For CAT, maximum decrease was found at 96 h of duration for both the concentrations, which was 63.37% for highest sublethal dose and 33.13% for lowest sublethal dose. However, the increase in values of CAT and SOD was observed after 720 h. SOD is involved in the breakdown of superoxide radicals to molecular oxygen and hydrogen peroxide. Hydrogen peroxide is further partitioned by catalase into water. As the inhibition of CAT by superoxide radicals have been reported by Kono and Fridovich [33]. The decrease activity of enzymes observed in the present study might be due to excessive generation of superoxide radicals, which may lead

to the decrease in activity of SOD and thus CAT. Almashhedy [34] also revealed the impact of textile dyes on antioxidant enzymes showing alteration in their activities in textile workers. The free radicals generated interaction with biomolecules such as DNA resulting in the formation of DNA adducts, which prevents replication accurately. As DNA carries genetic information, its stability and integrity is crucial for life sustainability. So, the damage associated with DNA caused by 2NS needs scientific approaches to combat the problems.

In the present study, genotoxicity of 2NS in *C. punctatus* was evaluated through micronucleus and comet assay. Maximum micronucleated cell frequency was observed at 240 h of exposure, which comes to be 1.63 ± 0.02 at highest sublethal dose. Maximum increase in tail length and % tail intensity was observed after treatment with both concentrations of 2NS as compared to control, which indicates the genotoxicity inducing potential of 2NS. Similarly, Fernandes et al. [35] revealed the toxicity induced by textile dye C.I. disperse blue on bone marrow of rats. The investigation of all parameters shows evident recovery within 720 h (30 days). The reason might be the presence of sulphonic group, which makes the compound hydrophilic in nature and become able to be eliminated from the body of the organism. The other reasons may be the activation of gene like cytochrome p450 [36] or may be removal of apoptotic cells with time [37]. Similar study performed by Sharma and Chadha [38] also showed the recovery gained by fresh water fish *C. punctatus* after the cessation of exposure to non-ionic surfactant, nonylphenol.

Bioaccumulation of xenobiotic said to elevate ROS, thus creating imbalance between the internal stress combat systems of the fish leading to the generation of oxidative stress. The bioaccumulation potential of 2NS in blood plasma of *C. punctatus* after treatment with highest dose was also revealed in the present study. The result showed the decrease in concentration with time. The bioaccumulation studies are consistent with toxicity studies as 2NS is found to be eliminated from the body of the organism and maximum concentration is found at 24 h. Similarly, the quantification of clenbuterol in bovine liver samples was done by Trejo et al. [38] using HPLC.

Conclusion

To the best of our knowledge, this is the first study revealing the toxic effects of 2NS after treatment with sublethal concentrations. The study presents the bioaccumulation, genotoxic, and oxidative stress inducing potential of 2NS—a toxic intermediate compound used in textile industries. This study could be used as solid foundation for the investigation of methodologies involved in the breakdown of 2NS either physically, chemically, or biologically.

Conflict of interest statement

The authors declare that they have no conflicts of interest.

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